2025 Home Energy Savings - Retrofits

Installation Contractor

Program Manual

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2025 Home Energy Savings - Retrofits Installation Contractor Program Manual

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Introduction

The Home Energy Savings - Retrofits Program ("Program") follows established program policies and provides income eligible customers with electric- and gas-saving retrofits at no cost. The program serves ComEd, Nicor Gas, Peoples Gas, and North Shore Gas ("the Utilities") customers in single family properties (defined as properties with fewer than 3 dwelling units); customers must have an active account with one of the Utilities and qualifying household income. The Program is implemented by Resource Innovations ("Implementation Contractor"). The Program's Installation Contractors each act as the customer's primary point of contact during the project, in collaboration with the Company (Resource Innovations), the Utilities, and community-based partners, as needed.

The following document outlines the policies and procedures applicable to the Installation Contractors' activity in the field, including customer outreach and engagement, customer intake and enrollment, initial property assessment, installation of comprehensive retrofits, and the post-installation quality control requirements. This manual may be amended during implementation.

Program Contacts

Resource Innovations HES-Retrofits Program Manager

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1. Installation Contractor Requirements

1.1 Preparedness Checklist

Please see below for the Installation Contractor preparedness checklist, completed for each Installation Contractor prior to commencing field activity.

Installation Contractor Preparedness Checklist

Initiative: Home Energy Savings - Retrofits
Installation Contractor:
Installation Contractor Contact:
Resource Innovations ("RI") Program Manager: Anthony Villegas

Completed (Check box)	Task	RI PM Initials	Installer Initials
	Certificate of Insurance issued by Installation Contractor and approved with appropriate limits and additional insured requirements per the Contract		
	W-9 provided to the Resource Innovations' Accounting Team		
	Installation Contractor Operations, Safety & Quality Plans submitted & approved		
	Installation Contractor reporting templates provided by RI		
	Issue, Elevation and Complaint Resolution Process communicated, outlined and approved		
	Installation guidelines and other program operations (e.g., assessment / installation / customer outreach processes, procedures, expectations, QAQC processes, etc.) delivered		
	Installation Contractor contact list provided with emergency & backup contacts identified		
	Weekly (or similar) coordination meetings established		
	RI approval provided to commence field-based work		

1.2 Personal Protective Equipment (PPE) guidelines

The following items are considered PPE and should be worn by Installation Contractor field staff when conducting assessment and installation services in customer homes.

*Exceptions may apply.

- **Safety Glasses:**
  - Standards:

        - Wrap-around style with a clear lens, including side shields
        - Lens meets or exceeds ANSI Z87.1 safety standards
    - Worn by field staff at all times while in a customer home conducting installation and select assessment tasks
    - *May be removed when safe to do so in order to clean lens
- **Safety Gloves:**
    - Standards:
        - Cut-resistant
        - Meet or exceed ANSI/ISEA 105-2005 performance ratings in the following areas:
            - Abrasion Resistance level - equal to or greater than level 4
            - Cut Resistance level - equal to or greater than level 2
            - Puncture Resistance - equal to or greater than level 2
    - Always Worn by field staff while installing measures
    - *May be removed when safe to do so during the installation of select measures, including programmable thermostats or the re-programming of existing programmable thermostats

ANSI/ISEA 105-2005 Mechanical Ratings:							
Rating	**Level 0**	**Level 1**	**Level 2**	**Level 3**	**Level 4**	**Level 5**	**Level 6**
Abrasion Resistance* (Cycles)	< 100	≥ 100	≥ 500	≥ 1000	≥ 3000	≥ 10000	≥ 20000
Cut Resistance (Grams)**	< 200	≥ 200	≥ 500	≥ 1000	≥ 1500	≥ 3500	-
Puncture Resistance (Newtons)	< 10	≥ 10	≥ 20	≥ 60	≥ 100	≥ 150	-
* Abrasion ratings 0 through 3 are based on measurements with a 500-gram load. Levels 4 through 6 are measured with a 1,000-gram load. **Weight needed to cut through material with 25mm of blade travel.							

- **Footwear:**
    - Standards:
        - Work-style boot or shoe with laces fully tied
        - Brown, black, or tan in color
        - Contain a minimum ½" raised heel
        - Sole must be non-marking and skid-resistant
        - Tennis shoes are not allowed
    - Always Worn by field staff while in a customer home and other work-related areas (e.g., warehouse, in and around property)

- **Protective Footwear Coverings:**
    - Standards:
        - Equal to or greater than 5 mil thick
        - Resist grease and liquids
    - Always Worn by field staff, unless a specific safety issue is present, including but not limited to:
        - Use on a ladder is not permitted, and should be removed prior to stepping onto a ladder
        - Slippery floor surfaces
    - In the event the field staff determines it is not safe to wear the shoe cover, field staff shall take every precaution to avoid damage to customer property when removing the shoe cover. Field staff are encouraged to wear "indoor-only" shoes to limit tracking of dirt into the home when shoe covers are not possible.

- **Bump Caps, Face Masks, Other Applicable PPE:**
    - Note: Installation Contractor will train all field staff on OSHA and BPI requirements applicable to entering the attic and other enclosed spaces, and provide the appropriate PPE including bump caps and face masks

- Field staff must wear bump caps and face masks, as required by OSHA, when entering an attic or other enclosed space

- **Ladder:**
  - Standards:
    - Legs should be fully extended and on a level and flat surface.
    - Ladders not in use should not be leaned against the owner's personnel property or any surfaces within the residence.
    - Ladders should be extended/opened and placed in a safe location while not in use.
    - Ladders will have a maximum weight capacity of 300lbs, or as applicable for the field staff.
  - Field staff must not use or borrow customer ladders.
  - Field staff must not allow customers to use program ladders.
  - In the event a ladder may not be properly set up, field staff must defer completion of the assessment or installation task.
  - Field staff may use a ladder attached to the property (e.g., attic hatch, other access) only after a thorough inspection of the ladder to ensure it is safe to do so.

1.3 Customer Service Expectations

The Program values all customers and participants and ensures a high level of customer satisfaction throughout the process. Installation Contractors are expected to make every effort to ensure a positive customer experience during all interactions with utility customers, participants, and the general public. Program and field staff are instructed to provide high quality customer service courtesies, respect customer property, be respectful during interactions, and present the utilities and the Program in a positive manner. Staff are not to engage in conflict or other unprofessional behavior with customers or the general public while operating under the Program. Staff should be provided basic training around deescalating conflict should conflict arise. If RI is made aware of any negative customer experience resulting from contractor behavior, via customer satisfaction surveys, QAQC findings, or otherwise, they will address the situation with the contractor and enforce any necessary corrective action, probation, or reconsideration of status in the program in accordance with the Participation Agreement and Program Manual.

In the event ~~Program~~ Installation Contractor staff are engaged by a customer due to a complaint or issue with the utilities, staff notify the customer that they are not directly affiliated with the utilities and refer the customer to the utility call centers:

ComEd

- Company inquiries: 800.334.7661
- Energy efficiency inquiries: 855.433.2700

Nicor Gas

- Company inquiries: 888.Nicor4U
- Energy efficiency inquiries: 877.886.4239

North Shore Gas

- Company inquiries: 844.889.3502
- Energy efficiency inquiries: 866.556.6004

Peoples Gas

- Company inquiries: 844.889.3501
- Energy efficiency inquiries: 866.556.6001

Program staff do not engage with the customer on any issue unrelated to the Program offerings.

1.4 Safety Protocols and Training

Safety is the top priority for the Program.

Company provides all staff and Installation Contractors with the safety training and protocols outlined in **Appendix A: Safety Plan.**

1.5 Field Safety

Safety is our top priority. Adhere to industry and Program safety standards while traveling to and from project sites and performing all work associated with the Program.

1.5.1 Handling of Incidents and Reporting

In case of field safety incident, reference Attachment A of the Contractor Participation Agreement.

Emergency situations (i.e., no heat due to the services provided by the Installation Contractor, fire hazard following measure install, etc.) and Level 3 complaints must be resolved immediately and within no more than 24 hours. RI must be notified of customer incidents within 1.5 hours. Level 1 and Level 2 issues do not require RI notification, but should be promptly addressed by Contractor.

1.5.2 Types of Reportable Incidents

Incident Levels

- Level 1 – Issue/concern/feedback directly provided to the program that can be resolved by the initial program contact
    - Customer upset regarding their high bill
    - Customer upset with the aerators or other measures being installed

- Level 2 – Issue/concern/feedback directly provided to the program in which the primary purpose is to notify of less than satisfactory experiences or opinions relating to any aspect of the program including, but not limited to, ~~property damage,~~ customer complaints, equipment failures, program offerings & rules, eligibility, timeliness, or customer service. Does not need further input for resolution.
    - Customer expresses dissatisfaction with the program and/or its offerings
    - Technician is late to the designated appointment window
    - Measure stops working within warranty

- Level 3 – Issues received from the client or applicable regulatory agencies or issues directly provided to the program that must be escalated due to the following reasons:
    - Property damage claim
    - Issue that could result in monetary compensation to the customer
    - Customer threating or asking for management and cannot be pacified
    - Repeated complaints from the same customer
    - Issue that could disrupt program operations or negatively impact the program's success or ability to reach its objectives
    - Issue that cannot be resolved by other team members
    - Issue that could cause irreparable damage to the client or program's name, brand, or public image
    - Issue that the program determines to have a strong likelihood of being reported to a regulatory agency or to the client directly
    - Any other issue that requires Company and Utility weigh-in for resolution

        - Technician breaks lamp while on-site
        - Technician notices the customer is very upset and feels threatened or sees weapons sitting out at the home
    - HVAC – No heat or no AC issue after Thermostat install
    - Vehicle and OSHA Accidents:
        - Must occur while working on the program
        - Regardless of who is at fault
    - Safety Hazards:
        - Fire hazard (identified during assessment, or caused during install)
        - Gas leak – Must contact Gas Utility
            - Nicor Gas Emergency (888.642.6748)
            - North Shore Gas Emergency (866.556.6005)
            - Peoples Gas Emergency (866.556.6002)
        - Elevated CO levels (levels above 35ppm require notification to property owner)
        - Any other concern about the customer's safety in relation to the work performed
        - No-heat for customer (when weather or other condition is a risk)

1.6 Invoicing and Data Support

Installation Contractor is expected to collect necessary data for savings calculations. Data must be input into the Resource Innovations Data Collection Tool ("DCT").

Installation Contractors will be provided a DCT Manual and webinar-style DCT training prior to entering the field for work, DCT manual will provide guidance on what data must be collected for eligible measures. Reference **Appendix C: Data Collection Tool Manual and Webinar Presentation**.

Installation Contractors are encouraged to use the **Energy Audit Data Collection Form (reference Appendix D)** as a guide to collect necessary data for project pre-approval. Systems information collected during a customer's home energy audit must be input into the DCT for RI pre-approval of any projects. Along with required data, a signed **Customer Authorization Form (reference Appendix E)** must be attached to all projects submitted for pre-approval.

Once a project is complete, data must be input into the DCT to submit projects for invoicing. Projects will be reviewed for final approval by RI. Contractor will utilize the **Documentation and Photo Checklist Form (reference Appendix J) to** ensure all required documentation is included. If critical data is missing, projects will be rejected and the Installation Contractor will need to remedy the missing documentation

Please ensure that photos are uploaded to the DCT in zip folders or a link to Company Cam or other web-based photo library is provided in the project. Also ensure that uploaded documents are named appropriately (e.g., the name of the document being uploaded with the address included), legible, and attached as complete documents (i.e., not attached as individual pages).

Project submittal deadlines and invoicing will follow the program calendar provided by RI.

1.7 Contractor Scorecard

Contractors will be provided a monthly scorecard by RI, to help facilitate performance management discussions (e.g., project volume relative to goal, forecasting accuracy, cost effectiveness, QAQC items, etc.) and identify areas for continuous improvement.

2. Program Eligibility and Benefits

2.1 Property Eligibility

Single-family properties, defined as properties with fewer than 3 dwelling units, are eligible for the program.

Homes requiring major repairs or remediation of environmental hazards to enable proper installation of weatherization measures may be denied service until issues are addressed by the homeowner, using dedicated health and safety resources of the Program, and/or referred to a third-party resource for assistance.

Additionally, a home may not be weatherized if the property is for sale, or in the process of being sold. If the Program discovers that the home is listed "For Sale" during the process of weatherization, all work should be stopped as is reasonable during the installation process, the homeowner should be notified, and the job should be closed out. The process should be well documented with narrative and/or digital photos.

A home that is in the process of legal foreclosure is not eligible for weatherization assistance. However, the client's receipt of a potential foreclosure or foreclosure notice is not the start of foreclosure proceedings/judicial foreclosure process and does not affect weatherization eligibility.

This is a jointly funded program by both ComEd and the gas utilities. Homes must have a ComEd account to receive any electric savings measures.

Any customer with existing electric resistance heat should be getting prioritized for a retrofit, unless they have obvious H&S conditions that would exceed the H&S limit. Homes with electric heating and/or electric water heating, ~~please consider the~~ should have replacement of electric-resistance appliances with heat pump appliances scoped. Refer to Section 4 for more information.

2.2 Customer Eligibility

Eligibility for participation is based on applicants having an active utility residential account and qualifying household income. Customers attest to their income eligibility during the intake process, managed by Resource Innovations or a designated partner. A single-family property is a property with fewer than 3 dwelling units. If the home is >4,500 sq ft, Resource Innovations will confirm income eligibility with the customer. A customer is eligible to participate in the Retrofits program every 15 years.

Customers will be prioritized for a Retrofit via the following Retrofits Prioritization criteria, which is determined during the UES initial walk-thru:

Home must have (all of A through D and at least one of E or F):

A. SF-type configuration
   - In-unit mechanical equipment
   - Customer has decision-making power over building envelope (building envelope can be addressed comprehensively); the attic and/or basement of the unit can be served without impacting any other units.

B. No obvious deferrable conditions.
   - No blatantly obvious issues that would likely exceed the per home health and safety funding available
   - Current list is roof leaks, vermiculite, hoarding, active water damage/bulk moisture, cracked ceilings that prohibit attic insulation, knob and tube wiring, or severe disrepair of access points (e.g., lack of railings, deteriorating porch).

C. Attic air sealing and insulation opportunity.
   - Attic insulation must be less than 4 inches on average across the entire attic.
     - If attic does not have a hatch, criterion is marked "yes."
     - If attic is not accessible due to other physical constraints, the customer is asked if the attic is insulated. If they say it is, criterion is marked "no." If they say "I don't know" or "no," this criterion is waived (marked n/a).
     - If the home does not have an attic, this criterion is waived (marked n/a).
     - If customer refuses access to space, criterion is marked "no."

D. Basement/crawlspace air sealing opportunity.
   - At least 20% of rim joist is accessible and not air sealed.
     - If the home does not have a basement, this criterion is waived (marked n/a).
     - If basement is finished (i.e., rim joist and ducts are not accessible) and home does not have a crawlspace, criterion is marked "no."
     - If basement is finished and home has a crawlspace, criterion is marked "yes."
     - If there is only a crawlspace, criterion is marked "yes."

E. Duct sealing opportunity.
   - Ducts located in semi-or unconditioned spaces.

F. HVAC equipment is inefficient.
   - Existing furnace or boiler is < or = 80% efficient and/or AC is < or = SEER 13

OR home must have:

G. An electric resistance heating system
   - Home must also meet A and B above

3. Project Process

1. Once prioritized for a retrofit, Inquiry generated by RI or enrollment partner
2. Within 2 days, Installation Contractor contacts customer to schedule assessment
3. Customer signs authorization form
4. Contractor completes assessment. Staff performing assessment must be certified as a BPI Building Analyst Professional (BA-P).
   a. If customer is deferred, be sure to cite this under Customer Status in the data collection tool with an explanation of the deferrable conditions and complete the **Hazardous Conditions Form (Appendix I)**
      i. Contractor should present the completed form.to the customer following the assessment and collect a signature indicating that they understand the potentially hazardous conditions in the home.
   ~~a.~~ The signed form should be uploaded to the project in the DCT. If the customer refuses to sign the form, contractor must add a note in the project with the reason the form was not signed.
      ii.
      ~~i. Note: No customer should be deferred until the program agrees that the issues identified by the contractor cannot be mitigated.~~
   b. Contractor will inform customer via verbal and/or written communication the reason(s) why the project cannot proceed. A customer may be deferred due to many reasons, here are some examples: conditions in the home that must be remedied before work can commence, there are no energy efficiency opportunities, crew safety, etc. In these instances, the contractor will

explain to the customer that they can't be served. unless there is a deferrable condition that can be remedied by the customer to allow the project to proceed. ~~at this time/at all, and provide reasons verbally and/or in writing.~~

5. Customer agrees to scope of work proposed by Contractor
    a. Before work begins in the home, a scope of work should be provided to the customer in writing clearly listing the measures that will be performed in the home, and a copy signed by the customer should be uploaded to the DCT. If changes are made or additional measures are performed in the home, an updated or additional scope of work should be created and signed by the customer.
        i. If contractor is performing additional work for a customer outside the program's approved measures, a separate scope of work must be signed to make clear it is not the responsibility of the HES program to pay for this additional work.
6. Contractor installs approved measures
    a. Customer needs to acknowledge that work is complete by signing the **Final Inspection Form (Appendix F)**.
7. Contractor inspects work to ensure customer is left in safe conditions and measures installed properly. Staff performing final inspection must be certified as a BPI Building Analyst Professional (BA-P).
    - CMC conducts QA/QC inspections on 5% of projects
    - After final inspection and customer education, Contractor should provide customer satisfaction survey for customer to mail in to Program. See **Appendix K: Customer Satisfaction Survey.**

Full details of this process are outlined in the **HES Steps of Service Field Manual (Appendix B).**

3.1 Assessment Scheduling

RI will notify Installation Contractors of eligible customers for assessment via the DCT. Installation Contractors must contact to schedule an appointment with the customer within 2 business days of notification from RI. Installation Contractor will then update the database (Powerpath) with an assessment date.

3.2 Assessment Procedures

Each Installation Contractor will conduct the following steps during the assessment.

The **HES Steps of Service Field Manual (Appendix B)** outlines the entire assessment process for the installation contractor.

The steps below provide a general outline for conducting the assessment.

3.2.1 Assessment and Diagnostic Testing:

1. Specific data points and diagnostic testing requirements are outlined in **Appendix B**.

3.2.2 Health and Safety:

Installation Contractor will continually monitor health and safety issues and follow the most current BPI practice standards to address any immediate health and safety concerns.

If any measured concentrations of combustible fuel gas exceed 10% of the lower explosive limit (LEL), the Installation Contractor shall inform the homeowner/occupants of the unsafe condition and advise evacuation of the home. After evacuation, Installation Contractor should call the gas utility emergency line; Nicor Gas (888.642.6748), North Shore Gas (866.556.6005), Peoples Gas (866.556.6002). Call outside of the home for gas leak-related issues, and inform Resource Innovations, who will notify the utilities Program Managers immediately.

3.2.3 Project Scoping:

The Energy Assessment will seek opportunities for three (3) measure categories:

1. Weatherization measures
2. Mechanicals – Mechanical repair, tune-up or replacement must meet decision tree criteria outlined in Appendix H: Mechanical Replacement Decision Trees ~~standards outlined in the IL TRM~~
3. Health & Safety Measures

3.2.4 Data Collection

Company will provide Installation Contractor with a data collection tool (reference **Appendix D: Energy Audit Data Collection Form**) for customers' assessment appointment. This tool should be utilized during an assessment to ensure all the necessary data for project savings calculations are accurately collected. A list of example data points can be found below:

a. Existing baseline conditions for each eligible measure
b. Photos of each eligible measure's pre-existing condition are required
c. General characteristics of home:
    i. Number of occupants
    ii. Square footage of home
    iii. Number of stories
    iv. Year Built
    v. Home Type
d. If larger building shell retrofits (i.e., attic insulation, air sealing, etc.) are not recommended for the customer's home:
    i. Assessor must make a note on the customer energy audit form. This note must then be recorded in RI's system under the "measure notes" section of the Assessment measure.

If a measure that was scoped to be installed is no longer going to be included with the project, all documentation needs updated, the DCT needs to be corrected to reflect the work that will be completed, and the customer should sign the updated scope of work.

If a measure that was scoped, installed and paid needs to be removed, the Installation Contractor will reach out to the Company to inform them of the change. This communication should include pictures of the item(s) removed. Company will make adjustments to future incentive payments to reconcile.

3.3 Quality Assurance

The Installation Contractor is responsible for conducting on-site final inspections. Final inspections should ensure all eligible measures have been installed, and work is performed in accordance with Program guidelines using the Program-provided **Final Inspection Form (Appendix F)**. CMC (subcontracted to RI) will conduct inspections in addition to the Installation Contractor's required inspections at a rate of no less than 5% of projects completed using a Program-provided **QAQC Observation Form (Appendix G)**. Company reserves the right to QAQC a contractor at elevated levels (above 5%) based on probation statuses defined below, in addition to other trends, findings, or for any reason deemed necessary.

Installation Contractor is responsible for resolving all project inspection, customer service, workmanship and warranty issues in a timely manner and for promptly notifying Implementation Contractor of all such issues discovered or reported to Contractor and subsequent resolutions.

If QAQC (by contractor or CMC) results in concerns for customer health and safety, the identified issues must be addressed by the contractor. Contractor commits to sharing a plan to resolve the issue within 24 hours and

resolving the issue within the week. If delays occur, contractor commits to communicating causes for delay and a timeline for resolution within a week of the issue being identified.

New Installation Contractors will enter the program under a probationary status to ensure that they have proper understanding and implementation of program guidance. In this initial probation period, they will have their first 10 completed projects QAQC'd by CMC, with the expectation that at least 8 of the 10 projects (80%) are completed in such a manner that no corrective action is required (i.e., they "pass" the QAQC). Corrective action means that the contractor needs to return to the home to correct a portion of the initial installation that was identified to not be in compliance with program standards. Should an Installation Contractor not be able to pass 8 of the 10 projects, the Installation Contractor's status in the program is subject to review and reconsideration.

Once an Installation Contractor has been established within the program, persistent QAQC issues may also trigger a probation status. If an Installation Contractor has completed at least 5 projects in the program year and their QAQC pass rate (number of projects that don't require corrective action/total number of projects) dips below 80%, they will be put on probation. The pass rate will reset at the beginning of each program year. The severity of individual QAQC findings or other incidents outside of QAQC may also cause a contractor to be immediately placed on probation, at the Company's discretion (e.g., finding that a customer was left in a very unsafe or dangerous situation, learning of property damage resulting from faulty workmanship and/or not adhering to program guidance, etc.). While on probation, the Installation Contractor will have their next 10 completed projects QAQC'd by CMC, with the expectation that at least 8 of the 10 projects (80%) are completed in such a manner that no corrective action is required (i.e., they "pass" the QAQC). Corrective action means that the contractor needs to return to the home to correct a portion of the initial installation that was identified to not be in compliance with program standards. The timeline to complete 10 projects will depend on contractor capacity and Company discretion. Should an Installation Contractor not be able to pass 8 of the 10 projects, the Installation Contractor's status in the program is subject to review and reconsideration.

~~An Installation Contractor newly joining the program will have 100% of their jobs QAQCed by CMC to ensure proper understanding and implementation of program guidance. The expectation is that at least 10 projects of the first 15 will be completed in such a manner that no corrective action is required in the customer's home. Should an Installation Contractor be unable to meet this standard, the Installation Contractor's status in the program is subject to review and reconsideration.~~

~~Once an Installation Contractor has been established within the program, persistent QAQC issues might trigger a return to the QAQC requirements for new Installation Contractors. If an Installation Contractor has three consecutive projects that do not pass QAQC (i.e., require corrective action in the customer's home), 100% QAQC of all projects will resume with the expectation that at least 10 projects of the next 15 will be completed in such a manner that no corrective action is required in the customer's home. Should an Installation Contractor be unable to meet this standard, the Installation Contractor's status in the program is subject to review and reconsideration.~~

Please note that QAQC can bring to light a variety of issues with Installation Contractor performance (e.g., customer satisfaction, accurate documentation, and reporting, etc.). While the above outlined supportive QAQC process is intended to aid the Installation Contractor in continuous improvement, the Contractor Non-Compliance Section of the Contractor Participation Agreement is always still in effect and can supersede the process outlined above.

~~3.4 Quality Assurance Protocol~~

~~Installation Contractor is responsible, with input from Resource Innovations and the utilities, for designing and implementing a QA/QC plan that will focus on program design and implementation.~~

~~Key areas of field operations QA/QC include but are not limited to:~~

- ~~Safety~~
- ~~Appearance~~
- ~~Assessment procedures~~
- ~~Installation procedures~~
- ~~Care of customer property~~
- ~~Data documentation and reporting~~

~~Any identified issues will be reported by Installation Contractor to Resource Innovations and the utilities by Company within 24 hours of identification, with the exception of Level 3 incidents, which should be reported immediately.~~

3.5 Contractor Warranty Requirement

Installation Contractor is responsible for providing a warranty on all work performed in a customer's home.

For new equipment installed in the home (thermostat, heating system, air conditioner, etc.), a one-year warranty shall be provided for all labor and equipment. Manufacturers' product warranty information and maintenance instructions shall be provided to the customer once equipment has been installed. For other installed measures, such as insulation and air sealing, contractor shall provide a one-year warranty on labor and installed material.

4. Eligible Energy Efficiency Measures

Approved energy efficiency measures provided by the Program are listed in the table below. All installations must follow BPI Standard Work Specifications, NREL Standard Work Specifications, local codes and manufacture recommendations. Specifications in table below supersede material specifications noted in above references. Additional installation requirements are outlined in **Section 6: Measure Eligibility Requirements**.

Please note, the **building permit fee** associated with the job is an allowable expense. It should be included in the list of measures in the DCT (Powerpath), and it should be input as a Health & Safety measure.

4.1 Single Family Measures

Measure	Notes
Assessment Fee	
Single Family Assessment	
Direct Install Measures	
Programmable Thermostat	Advanced thermostat is preferred, however, if customer requests a programmable one instead then it is installed. Thermostat must have capability to adjust temp set points according to 7 day schedule set by the installer in collaboration with the homeowner. Existing thermostat must be functional in order to be replaced.
Advanced Thermostat	Advanced thermostats may be installed to replace manual or programmable existing thermostats for homes with a furnace and/or CAC. Homes with an existing boiler should only have an advanced thermostat installed if the existing one is in poor condition. Existing thermostat must be functional in order to be replaced.
Insulation Measures	
Attic Insulation	Attic insulation should be installed when existing attic insulation is rated at R-19 or lower. Existing insulation levels should be brought up to R-49.
Basement/Crawlspace Wall Insulation	Basement and/or crawlspace walls may be insulated if there is no existing insulation. New insulation should be minimum R-10.
Floor Insulation Above Crawlspace	Floor above an unconditioned crawlspace may be insulated if there is no existing insulation. New insulation should be a minimum R-20. If ductwork is within the crawlspace, then proper thermal boundary would be the crawl space walls.
Wall insulation	Walls may be insulated if there is no existing wall insulation or if the existing insulation is in poor condition.

Low-e storm windows	~~Replacing missing or broken glass~~Storm window can be added to inoperable single pane windows.
AC Covers	A rigid cover that covers the indoor side of the air conditioner unit with foam gaskets to seal the edges OR A flexible cover that is well insulated and perfectly covers the indoor side of the air conditioner unit
Rim Joist Insulation	Rim/band joist may be insulated if there is no existing insulation. New insulation should be a minimum of R-10.
Air Sealing and Duct Sealing	
Air Sealing	Air sealing should be performed on all homes, with a goal of an overall reduction in CFM50 of 25% or greater. Homes must have a CFM50 reading ≥110% of the building's square footage.
Duct Sealing	Eligible duct systems are located in unconditioned and/or semi-conditioned spaces. Duct work must have CFM25 pre/post readings.
Mechanicals	
Gas Furnace Replacement	Emergency replacement only. See **Mechanical Replacement Decision Trees (Appendix H)** for baseline conditions that allow for repair vs replacement. Efficiency of new furnace must be ≥95%
Electric Furnace Replacement	Electric resistance furnaces can be replaced with ASHPs, regardless of age or condition. Please ensure the home's heating load can be served by the ASHP. New ASHPs must meet NEEP Cold Climate specifications: • HSPF2 ≥ 7.7 (9 HSPF equivalent)~~HSPF ≥ 9~~ • COP @ 5° F ≥ 1.75 • SEER2 ≥ 14.3 (15 SEER equivalent)~~SEER ≥ 15~~ Please consult with the homeowner, and gain their approval, about replacement of furnace if existing furnace has been replaced recently.
Boiler Replacement	Emergency replacement only. See **Mechanical Replacement Decision Trees (Appendix H)** for baseline conditions that allow for repair vs replacement. New boiler efficiency should be ≥ 95%.
Electric Resistance Base Board Replacement	Electric resistance base board can be replaced with ductless mini splits, regardless of age or condition. Please ensure the home's heating load can be served by the ductless mini splits. New ductless mini splits must meet NEEP Cold Climate specifications: • HSPF2 ≥ 8.5 (10 HSPF equivalent)~~HSPF ≥ 10~~ • COP @ 5° F ≥ 1.75 • SEER2 ≥ 15 (15 SEER equivalent) ~~SEER ≥ 15~~

Natural Gas Water Heater Replacement	Emergency replacement only. See **Mechanical Replacement Decision Trees (Appendix H)** for baseline conditions that allow for repair vs replacement. Energy factor of new water heater must be ≥ 0.67.
Electric Resistance Water Heaters Replacement	Existing electric resistance water heaters can be replaced with Heat Pump Water Heaters regardless of age or condition. New Heat Pump Water Heater must be Energy Star Rated. Please consult with homeowner, and gain their approval, about replacement of water heater if existing water heater has been replaced recently.
Central Air Conditioner Replacement	Emergency replacement only. CAC must be ≤SEER 10 and manufactured before 2000 to qualify for replacement. The efficiency of the new CAC must have SEER 2 ≥ 15.2 (16 SEER equivalent). See **Mechanical Replacement Decision Trees (Appendix H)** for baseline conditions that allow for repair vs replacement.
Gas Furnace Tune-Up	Furnaces must not have received a tune-up within the last 3 years. Note: Electric furnace tune-ups are not allowed as the furnace should be replaced with a heat pump.
Boiler Tune-Up	Boilers must not have received a tune-up within the last 3 years.
Room Air Conditioner	Emergency replacement only (i.e., room A/C is nonfunctional). Replace with Energy Star efficiency.
EC Motors	When furnace or HVAC system is older but running well Static pressure should be taken to ensure system is functioning as specified

Standard Health and Safety Measures

Health and Safety Measures cannot exceed $1,500 per home. Exhaust fans and mechanical replacement are not to be included in the cap; those measures are separate from the health and safety measures cap. Mechanical repairs are to be included in the $1,500 cap; tune-ups are not.

Health and Safety Measures	
Kitchen Exhaust Fan	Baseline equipment = existing bath/kitchen exhaust fan is in poor condition. New equipment = sized per ASHRAE 62.2 standards. A screenshot of the fan RedCalc is required to be uploaded to the project in Powerpath. Make sure to also include basement area in the ASHRAE calculation. Include fan flow rates on the assessment report.
Bathroom Exhaust Fan	
Bathroom Exhaust Fan w/ Light	
Smoke Detector – Change Out	Smoke and CO detectors should be installed according to local code requirements. Must use a 10-year sealed battery and replace any battery-operated smoke detector older than 10-years in accordance with new law in 2023.

Smoke Detector – Hardwired	All smoke alarms shall be listed in accordance with UL 217 and installed in accordance with the provisions of this code.
CO Detector	CO Detectors must meet Standard UL 2034 and shall be installed in accordance with the above and the manufacturer's installation instructions.
CO Detector Hardwired	
CO/Smoke Detector Combination – Hardwired	
9 Volt Battery Replacement	Installed if existing battery on a CO detector is not functioning.
Dryer Vent Pipe	Exhaust vents on dryer or combustion appliance is improperly installed or missing. All vents must terminate outside the building shell and never in an attic. At a minimum, semi-rigid dryer duct is to be used for new dryer vent.
Dryer Vent Termination	
Battery Replacement – AA or AAA	Installed if existing battery on a CO detector is not functioning.
Exhaust Vent Termination	If not associated with the installation of a fan.
Gas Mechanical Repairs	Existing gas furnace, boiler or water heater does not meet replacement guidelines and repairs are necessary to resolve an operational issue or remedy a health and safety issue.
Miscellaneous	Any health and safety items that fall outside the listed items above may be considered for installation/improvement, on a case-by-case basis. Home should not be deferred without proposing all health and safety measures needed to mitigate the deferral to the Program. Measures can be listed with associated costs in the DCT.

5. Health and Safety Standards

Installation Contractor will identify any Health and Safety issues during the Assessment phase. Should Health and Safety Measures total more than $1500.00 (not including exhaust fans or mechanical replacements) or $1000 for gas-only projects, The Company Program Manager will communicate with the utilities for approval of measures or referral to another program. A gas leak or ambient carbon monoxide reading above 35 PPM should trigger immediate action. In Nicor Gas Territory, a bubble solution test should be performed to confirm the site of a gas leak. If a gas leak or carbon monoxide issue is expected, Installation Contractor should evacuate everyone from the home and call the appropriate gas utility immediately.

Health and Safety Measures include, but are not limited to:

- Installation of Smoke Detectors and Carbon Monoxide detectors;
- Installation/Repair of exhaust fans in kitchens and bathrooms in accordance with ASHRAE 62.2 protocols;
- Correcting/Repairing unsafe or ineffective dryer venting;
- Correcting/Repairing leaking fuel supply lines;
- Correcting/Repairing improper or ineffective HVAC venting (such as installing a chimney liner);
- Remediation of conditions that may lead to or promote biological concerns and unsanitary conditions;
- Electrical repairs/upgrades necessary for weatherization measures and where the health and safety of the occupant is at risk;
- Minor correction of moisture and mold creating conditions when necessary to ensure the long-term stability and durability of the weatherization measures and the clients' long-term health and safety; If

greater than 10 square feet of mold is found localized in the home, remediation efforts and correcting moisture source must occur before air sealing can proceed.

- Gutter or downspout work when necessary to promote positive site drainage and to stop or prevent moisture/mold mildew conditions;
- Sump Pump repair, replacement, installation, and/or covers necessary to keep seepage water out of the dwelling to stop or prevent moisture/mold mildew conditions;
- Sealing of foundation cracks, the installation vapor barriers, sump pump covers to lower the risk of radon to occupants;
- Repair stairs and/or replacing hand railings for workers (and occupants) health/safety;
- Pest Removal;
- RRP lead safe work-related activities;
- Oven/Stove repair if a CO issue is noted (note: replacement may be justified if the cost of repair is equal to or exceeds the cost of replacement);
- If asbestos is suspected, discretion must be used to ensure customer safety.

~~Where vermiculite is present, the program requires mitigating the vermiculite before other measures can proceed.~~
Where vermiculite is present, treat as if asbestos-like material. Do not disturb. Do not use blower door.

Measures can proceed if asbestos-like material will not be disturbed (e.g., if there is asbestos-like pipe wrap in the basement, attic insulation could proceed). Special consideration should be given to the location of the asbestos-like material.

Blower door tests may not be conducted where asbestos-like material is found; however, air sealing, duct sealing, and insulation measures may proceed so long as an ASHRAE fan is installed and properly ventilating the home.

~~Where vermiculite is present, the program requires mitigating the vermiculite before other measures can proceed.~~

If you have questions about a certain situation, please escalate it to the Resource Innovations team for discussion.

6. Measure Installation Standards

6.1 Energy-Saving Measure Installation Requirements:

The following section outlines measure installation requirements, which are in addition to the measure-level eligibility and minimum efficiency guidelines outlined in Section 4.

- Programmable Thermostat – New Installation
  - Power is shut off completely at the furnace using the furnace shutoff switch.
  - The existing manual thermostat is uninstalled. Care is taken to avoid peeling paint or drywall in the removal of an old thermostat. Any mercury-containing thermostat is disposed of using the protocols listed in this document.
  - The new programmable thermostat is installed per manufacturer specifications.

- After installation, the furnace and air conditioner, if applicable, are tested for functionality, dependent on outdoor air temperature (note: the central air conditioner is not tested if the outdoor air temperature is 65 degrees Fahrenheit or lower).
- The new programmable thermostat is programmed with input from the customer, if available. The customer is provided with a brief tutorial of basic functionality, as well as on energy saving tips including using ENERGY STAR-recommended settings and setback guidelines. All user manuals and materials are left with the customer.
- If applicable, a backplate is installed to cover up any paint changes left by removing the original thermostat. Customer input is requested if the customer is available.

- Advanced Thermostat – New Installation
  - Prior to installation, the customer is screened as a candidate for an advanced / smart thermostat. Characteristics and requirements include, but are not limited to:
    - Customer has a smart phone or is comfortable with the advanced thermostat technology.
      - Note: A smart phone is not required as the thermostat senses home/away occupancy without requiring a smart phone interface. Additionally, the thermostat may be controlled solely at the device and does not have to be accessed remotely.
    - Customer has functioning Wi-Fi in their home or understands the limitations of the device without a Wi-Fi connection.
      - Note: Wi-Fi is not required for installation as the thermostat senses home/away occupancy without requiring an internet connection, and all settings may be adjusted at the device. The primary limitations of using an advanced thermostat without a Wi-Fi connection are limited access to energy usage data, inability to use real-time outdoor temperature readings from a weather service and inability to update the software.
      - Customers are encouraged to use the device with a Wi-Fi connection, however, lack of access to Wi-Fi will not disqualify a customer from receiving the device.
    - The existing heating and cooling systems, if applicable, are compatible with the device.
    - Customer is interested in and willing to use the advanced thermostat technology over a standard programmable thermostat.
  - Power is shut off completely at the furnace using the furnace shutoff switch.
  - The existing manual thermostat is uninstalled. Care is taken to avoid peeling paint or drywall in the removal of an old thermostat. Any mercury-containing thermostat is disposed of using the protocols listed in this document.
  - The new advanced thermostat is installed per manufacturer specifications.
  - After installation, the furnace and air conditioner, if applicable, are tested for functionality, dependent on outdoor air temperature (note: the central air conditioner is not tested if the outdoor air temperature is 65 degrees Fahrenheit or lower).
  - The new advanced thermostat is programmed with input from the customer. The customer is provided with a brief tutorial of basic functionality, as well as on energy saving tips including using ENERGY STAR-recommended temperature settings and setback guidelines. All user manuals and materials are left with the customer.
  - If applicable, a backplate is installed to cover up any paint changes left by removing the original thermostat. Customer input is requested if the customer is available.
- Attic Insulation
  - Attic insulation will be offered to all customers with an existing insulation of R-19 or less.

- Attics must be free of clutter and must be accessible by the assessor and Installation Contractor. A temporary attic hatch may be installed for the duration of the project to access the attic.
- Health and safety repairs or issues must be completed or resolved prior to proceeding with the project, as outlined in this document.
- Attic insulation is installed according to BPI specifications by a qualified contractor.
- Attic insulation should not occur in homes where air sealing is not being completed
- Chimney dams must be installed whenever doing attic insulation and insulation is in contact with chimney
- When installing ceiling insulation, please ensure the ceiling can support the added weight of the insulation by doing the following:
  - Visually inspect for cracks or other damage that may compromise the integrity of the drywall
    - Do not proceed with insulation until cracks or other damage is fixed
  - Visually inspect and use a level or other straight-edge tool to check for ceiling sagging or bowing; also inspect from attic and determine if there are gaps between drywall and joists, indicating sagging or bowing
    - Do not proceed with insulation until any sagging or bowing is fixed
  - Visually inspect (looking for nail pops or other visual indication) and use a magnet to check for drywall fastener type (if fasteners are set in pairs, they are likely nails, which may not hold with the added weight of insulation)
    - Do not proceed with insulation if nails are found until you fasten the drywall with screws
  - Make sure joists are spaced 16 inches or less on center (any wider joist spacing may not support the added weight of insulation)
    - Do not proceed with insulation if joist spacing is greater than 16 inches on center. Proceed with air sealing if deemed safe to do so.
- Note the condition of the ceiling finish before blowing cellulose. Cellulose weighs about twice that of blown fiberglass for the same R-value. If the ceiling condition is poor, consider using fiberglass.
- When installing ceiling insulation, please consider the impact of ceiling fans that may not be secure, which could fatigue or loosen drywall and compromise its ability to hold the added weight of the insulation.
  - Inspect ceiling fans to ensure they are securely attached to structural members above the drywall (i.e., not resting on the drywall). This should be done by the assessor at the time of the assessment and again by the crew/installers working in the area where a fan is installed.
  - Inspect from inside the attic, removing existing insulation from the area where the fan is located, and confirm
    - Light/fan fixture is attached securely and is not supported by the drywall
    - Structural members supporting the fan are not compromised in any way where fasteners attach (e.g., wood splitting, metal brackets ripped, etc.)
    - Fastener quantity, length, and type are adequate to keep fan secure

- Air Sealing (Infiltration Reduction)
  - Air sealing will be offered to all customers with an air leakage as measured at CFM50 greater than 110% of the square footage of the building/unit, per the diagnostic testing conducted during the assessment.
  - Air sealing will not be completed in a home or property where it is unsafe to do so, as determined by BPI. This includes, but is not limited to, a home or property with a CAZ issue, an

indoor air quality concern or where proper ventilation cannot be achieved, or a home or property with an occupant(s) that has a medical condition or other extenuating circumstance where air sealing cannot be done safely.
    - Air sealing is completed according to BPI specifications by a qualified contractor.
- Wall Insulation (Knee Wall)
    - The cavity must have no existing effective thermal resistance; insulate to R-11 or greater
    - This measure is intended for attic knee walls and attic spaces with sloped ceilings
- Rim Joist Insulation
    - Penetrations in rim joist must be sealed before insulating
    - Rim joint must have no existing insulation; insulate to R-10 or greater
- Duct Sealing
    - Duct sealing will be offered to all customers with a duct system that has existing ducts located in unconditioned and/or semi-conditioned spaces of the home.
        - A non-conditioned space is defined as a space outside of the thermal envelope of the building that is not intentionally heated for occupancy (crawl space, roof, attic, etc).
        - A semi-conditioned space is defined as a space within the thermal envelope that is not intentionally heated for occupancy (unfinished basement).
        - The existing baseline condition is a leaky duct work within the unconditioned or semi conditioned space in the home.
    - Duct sealing is conducted on all ducts within both the unconditioned and semi-conditioned spaces using mastic sealant.
    - A duct system that requires repairs may not be treated unless sufficient health and safety budget is available for the home to address minor repairs and complete the measure.
    - Duct work must have CFM25 pre/post readings. At a minimum, plenums, main ducts, takeoffs, and boots must be sealed.
        - Before sealing, measure total leakage at CFM25 using a duct blaster. Seal ducts using mastic, DuctEZ, tape, etc. After sealing, measure reduction at CFM25
- Furnace Tune-up (Natural gas furnaces only)
    - Existing furnace must not have had a tune-up within the last 3 years.
    - An existing furnace, determined not eligible for replacement, will be scheduled for a furnace tune-up by a qualified contractor.
    - Furnace tune-up must be completed according to American Standard Maintenance for Indoor Units as noted in the Illinois TRM and outlined in the Program Operations Manual.
    - The following tune-up services will be performed, as required by the IL TRM for savings calculation purposes:
        - Measure combustion efficiency using an electronic flue gas analyzer
        - Check and clean blower assembly and components per manufacturer's recommendations
        - Where applicable lubricate motor and inspect and replace fan belt if required
        - Inspect for gas leaks
        - Clean burner per manufacturer's recommendations and adjust as needed
        - Check ignition system and safety systems and clean and adjust as needed
        - Check and clean heat exchanger per manufacturer's recommendations
        - Inspect exhaust/flue for proper attachment and operation
        - Inspect control box, wiring and controls for proper connections and performance
        - Check air filter and clean or replace per manufacturer's recommendations
        - Inspect duct work connected to furnace for leaks or blockages
        - Measure temperature rise and adjust flow as needed
        - Check for correct line and load volts/amps

        - Check thermostat operation is per manufacturer's recommendations (or install new programmable or advanced thermostat, single family only, per Program guidelines)
        - Perform Carbon Monoxide test and adjust heating system until results are within standard industry acceptable limits

- Boiler tune-up
    - Existing boiler must not have had a tune-up within the last 3 years.
    - An existing boiler, determined not eligible for replacement, will be scheduled for a furnace tune-up by a qualified contractor.
    - Boiler tune-up must be completed according to the Act on Energy Commercial Technical Reference Manual No. 2010-4, 9.2.2 Gas Boiler Tune-up, as noted in the Illinois TRM and outlined in the Program Operations Manual.
    - The following tune-up services will be performed, as required by the IL TRM for savings calculation purposes (Note: Tune-up activities marked with an asterisk (*) are eligible to be performed at periods of boiler shutdown):
        - Measure combustion efficiency using an electronic flue gas analyzer
        - Adjust airflow and reduce excessive stack temperatures
        - Adjust burner and gas input, manual or motorized draft control
        - Check for proper venting
        - Complete visual inspection of system piping and insulation
        - Check safety controls
        - Check adequacy of combustion air intake
        - Clean fireside surfaces.
        - Inspect all refractory. Patch and wash coat as required.*
        - Inspect gaskets on front and rear doors and replace as necessary.*
        - Seal and close front and rear doors properly.*
        - Clean low and auxiliary low water cut-off controls, then re-install using new gaskets.*
        - Clean plugs in control piping.*
        - Remove all hand hole and manhole plates. Flush boiler with water to remove loose scale and sediment.*
        - Replace all hand hole and manhole plates with new gaskets.*
        - Open feedwater tank manway, inspect and clean as required. Replace manway plate with new gasket.*
        - Clean burner and burner pilot.
        - Check pilot electrode and adjust or replace.
        - Clean air damper and blower assembly.
        - Clean motor starter contacts and check operation.*
        - Make necessary adjustments to burner for proper combustion.
        - Perform all flame safeguard and safety trip checks.
        - Check all hand hole plates and manhole plates for leaks at normal operating temperatures and pressures.
        - Troubleshoot any boiler system problems as requested by on-site personnel

- Natural Gas Water Heater Replacement
    - New water heaters, when applicable, are installed by a qualified contractor to all manufacturer specifications, as well as local, regional and state codes.
    - Refer to the **Mechanical Replacement Decision Trees (Appendix H)** for guidance on replacing water heaters.
    - ~~Note on selection of a replacement over repairs:~~

- ~~Repairs may be suggested as a health and safety measure on any water heater that is not fully functional, if the cost of the repairs is less than 50% of the cost to install a new unit.~~
- ~~A full replacement will be recommended when the water heater:~~
  - ~~Is not functional at the time of assessment;~~
  - ~~Has an irreparable issue; *or*~~
  - ~~Is not providing sufficient hot water for the home and/or the occupants' needs, *and*~~
  - ~~The cost to repair exceeds 50% of a new replacement.~~
  - ~~Note: Installation Contractor may recommend a replacement, with justification, when the home and/or occupant would benefit from an immediate replacement rather than a repair. All recommendations are reviewed and approved by the Company Program Manager, and/or the gas utility (natural gas water heaters).~~

- Natural Gas Boiler Replacement
  - New boilers, when applicable, are installed by a qualified contractor to all manufacturer specifications, as well as local, regional and state codes.
  - Refer to the **Mechanical Replacement Decision Trees (Appendix H)** for guidance on replacing boilers.
  - ~~Note on selection of a replacement over repairs:~~
    - ~~Repairs may be suggested as a health and safety measure on any boiler that is not fully functional, if the cost of the repairs is less than 50% of the cost to install a new unit.~~
    - ~~A full replacement will be recommended when the boiler:~~
      - ~~Is not functional at the time of assessment;~~
      - ~~Has an irreparable issue; *or*~~
      - ~~Is not providing sufficient heat for the home and/or the occupants' needs, *and*~~
      - ~~The cost to repair exceeds 50% of a new replacement.~~
      - ~~Note: Installation Contractor may recommend a replacement, with justification, when the home and/or occupant would benefit from an immediate replacement rather than a repair. All recommendations are reviewed and approved by the Company Program Manager and the utilities.~~

- Natural Gas Furnace Replacement
  - New furnaces, when applicable, are installed by a qualified contractor to all manufacturer specifications, as well as local, regional and state codes.
  - Refer to the **Mechanical Replacement Decision Trees (Appendix H)** for guidance on replacing gas furnaces.

  - ~~Note on selection of a replacement over repairs or a tune-up:~~
    - ~~The furnace tune-up measure is available for all furnaces, regardless of condition, efficiency or age, if a tune-up has not been conducted within the last three years.~~
    - ~~Repairs may be suggested as a health and safety measure on any furnace that is not fully functional, if the cost of the repairs is less than 50% of the cost to install a new unit.~~
    - ~~A full replacement will be recommended when the furnace:~~
      - ~~Is not functional at the time of assessment;~~
      - ~~Has an irreparable issue; *or*~~

- ~~Is not providing sufficient heat for the home and/or the occupants' needs, *and*~~
- ~~The cost to repair exceeds 50% of the cost for a new replacement.~~
- ~~Note: Installation Contractor may recommend a replacement, with justification, when the home and/or occupant would benefit from an immediate replacement rather than a repair. All recommendations are reviewed and approved by the Company Program Manager and the utilities.~~

- Electric Water Heater Replacement
  - New Heat Pump water heaters, when applicable, are installed by a qualified contractor to all manufacturer specifications, as well as local, regional and state codes.
  - Note on selection of a replacement over repairs:
    - Replace all electric water heaters found, no matter of condition.
    - Consult with homeowner to gain their approval to replace existing water heater, if existing water heater has been installed recently by homeowner.
- Electric Resistant Base Board
  - Ductless mini splits, when applicable, are installed by a qualified contractor to all manufacturer specifications, as well as local, regional and state codes.
  - Notes on replacing electric baseboard heat with ductless mini splits:
    - ~~Replace~~ Remove all baseboard heat found, no matter of condition, except leave as much installed as is feasible with existing panel capacity (i.e., remove what you need from panel to support heat pump); prioritize leaving in wet areas that would want a backup heat source to keep pipes from freezing.~~.~~
    - Attach completed ACCA Manual J and Manual S calculations to project in the DCT
    - Refer to ComEd ASHP Training and specific manufacturer training for installation process
- Electric Furnace Replacement
  - New Air Source Heat Pump (ASHP), when applicable, are installed by a qualified contractor to all manufacturer specifications, as well as local, regional and state codes.
  - Notes on replacing electric furnaces with ASHPs:
    - Replace all electric furnaces found, no matter of condition.
    - Consult with homeowner to gain their approval to replace existing furnace, if existing furnace has been installed recently by homeowner.
    - Attach completed ACCA Manual J and Manual S calculations to project in the DCT
    - Refer to ComEd ASHP Training and specific manufacturer training for installation process
- AC Covers w/ Foam Gaskets
  - Must be installed on a room air conditioner (a window, sleeve, or through the wall air conditioner, PTAC, or PTHP) that is poorly installed with gaps around the edges. Installed cover or gap sealer must remain installed throughout the winter heating season and equipment must meet one of the following installation conditions to qualify:
    - Air conditioner unit remains installed in the window or sleeve throughout the year
      - A rigid cover that covers the indoor slide of the air conditioner unit with foam gaskets to seal the edges OR
      - A flexible cover that is well insulated and perfectly covers the indoor side of the air conditioner unit
    - Wall-thru or sleeve air conditioner unit is removed for the heating season
      - A rigid cover that fits inside the sleeve with foam gaskets along the edges for proper air sealing
- Low-e Storm Window Pilot
  - ~~Missing or broken glass~~Can be added to single pane, inoperable windows.

- Measure IGU in sash (W x H x Thickness)

- **EC Motors**
  - When system is older but running well
    - Static pressure should be taken to ensure system is functioning as specified

Use the **Documentation and Photo Checklist (Appendix J)** to ensure all items are included in order to submit each project for payment.

Cost Effectiveness

- To ensure Program is investing rate payer dollars responsibly, it is important to ensure the overall cost/savings ratio is within range.
  - Cost effectiveness will be communicated to Contractors quarterly to ensure costs are within acceptable range
    - Contractors should not have an average cost effectiveness more than double the network average for 3 or more months. If this occurs, there is the possibility for the Program to reallocate budget to contractors with better $/savings.

HES License and Certification Requirements for ~~Subcontractors~~Contractors

Contractors must follow local ordinances and obtain permits as required by local jurisdictions.

Comprehensive assessment or final inspection	Contractors performing the comprehensive assessment and/or final inspection of completed work must be certified as a BPI Building Analyst Professional (BA-P)
Roof repair or replacement	Must be a licensed roofing contractor recognized by the State of Illinois https://idfprapps.illinois.gov/Forms/Brochures/RooferBrochure.pdf
Plumber	Must be a licensed plumber by the State of Illinois DPHPlumbing@illinois.gov
Asbestos remediation	Must be licensed in the State of Illinois Department of Public Health
Mechanical repairs	No state license required; however, villages and municipalities may have licensing requirements. If working on AC, contractor performing work should have an EPA refrigeration certification
Electrical repair	No state license required; however, villages and municipalities may have licensing requirements Illinois Electrical License Requirements Guide (generalcontractorlicenseguide.com)
Sump pump installation	Must be a licensed plumber by the State of Illinois DPHPlumbing@illinois.gov
Heat pump installation (space heating)	Retrofits Contractors must complete the ComEd ASHP training associated with the ComEd Home Heating and Cooling Program (4 modules) and heat pump installers (Retrofits Contractor or subcontractor) must complete a heat pump manufacturer training.

6.2 Energy Savings Calculations

Company will use TRM-defined savings values to calculate all therms, kWh, and kW savings at the measure level. Where appropriate, Company will propose assumptions with rationale. Required data points used in savings calculations are captured during the assessment, installation and inspection, and will be included on the utility data file deliverables.

6.3 Handling, Transporting, Recycling, and Storage of Mercury Thermostats

Mercury containing thermostats that are removed by Program personnel must be taken from the unit and properly recycled. This is required by Illinois state law. All participating Installation Contractors are provided with proper recycling protocols and are instructed to dispose of mercury-containing thermostats at a collection site. Contractor will take the intact mercury thermostat, place in a sealed bag, then take to a hazardous waste facility. Note: This Program channel does not operate a collection site or a warehouse where a collection site could be located, found at: https://www.thermostat-recycle.org/zipsearch/. All thermostats are disposed of according to Illinois state law at a collection site.

To ensure proper handling and safety the following steps must be taken to handle, transport and recycle mercury thermostats:

- Each existing mercury containing thermostats is placed in its own separate Ziploc bag and sealed/zipped closed.
- Thermostats are placed inside a designated plastic bin inside the work van.
- Once the shift has ended all mercury containing thermostats must be removed from the plastic bag and placed into the recycling receptacle.
    - Only the piece that is attached to the mercury ampule should go inside the recycling bin per instructions provided by TRC.
    - All other pieces should be properly disposed of in the garbage.
- Instructions for proper contents and disposal will be clearly posted near the recycling bins.
- If for some reason the mercury recycling bin is not available, then the mercury containing thermostat(s) must remain inside the Ziploc bag and inside the plastic storage bin until the first opportunity to recycle the materials arises. Such products should not remain inside the work vehicle for more than 48 hours.

For reference, these requirements are contained in Public Act 096-1295 **"Mercury Thermostat Collection"** (http://www.ilga.gov/legislation/publicacts/fulltext.asp?Name=096-1295) with the key relevant statements below:

- Contractor: means a person engaged in the business of installation, service, or removal of heating, ventilation, and air-conditioning components
- No contractor or other person shall remove, replace, or otherwise take out of service a mercury thermostat unless the contractor or person delivers it to a collection site established under this Act
- Any thermostat wholesaler, contractor, or other person that violates any provision of this Act, or any rule adopted by the Agency pursuant to this Act, or that fails to perform any duty imposed by this Act shall be liable for a civil penalty not to exceed $500 per day for each violation. Each violation of this Act shall constitute a separate offense and violation. The penalties provided for in this Section may be recovered in a civil action brought in the name of the people of the State of Illinois by the State's Attorney of the county in which the violation occurred or by the Attorney General.

6.4 Bed Bug Protocol

Installation Contractor field staff are provided with the following protocols on how to properly recognize and identify bed bugs, and what to do if they are encountered during a site visit.

Bed Bug Facts:

- Bed bugs have a flat, oval-shaped body with no wings, and are 4-7mm long.
- Their color is shiny reddish-brown.
- People exposed to bed bugs may encounter bites that will appear as little red bumps on the skin, similar to a mosquito bite.
- There are currently no known cases of disease associated with bed bug bites.
- Bed bugs are mostly found in bedrooms or places where people sleep.
- They can be under chairs, couches, beds and dust covers.

Adults are small, brownish insects, just under a 1/4" long and are relatively flat. They are nearly as wide as they are long, and oval in shape. Immature bed bugs (nymphs) resemble adults but are much smaller and lighter in color. Newly hatched nymphs are translucent and are no bigger than a pinhead (1 mm). After feeding on a blood meal the immature bed bugs may appear bright red in color. Bed bugs lack wings and therefore they do not fly, but they are capable of moving swiftly on both horizontal and vertical surfaces. The eggs are very small (approximately 1mm), whitish, and very difficult to see on most surfaces without magnification (individual eggs are about the size of a dust speck). It is important to note that dirty living conditions do not attract bed bugs. Bed bugs can be found in the nicest hotels or homes.

Common Areas of Infestation:

Bed bugs are most commonly found in bed mattresses and bed frames because they are attracted to warm blooded mammals. They can be located anywhere in a home because bed bugs can thrive in so many different hiding places. What all of these places tend to have in common is the presence of folds, creases, seams, or crevices. These are ideal locations for bed bugs to lay their eggs; the bugs themselves may also adhere to these areas when not crawling forth at night to bite unsuspecting sleepers.

The most common locations that meet the criteria described above are beds and upholstered furniture, with the seams of mattresses being an overwhelmingly popular place for bed bugs to infest. However, the bug cannot thrive unless there is a host present to provide blood. This host may be a human being or any other warm-blooded mammal. Pets or other types of animals that may be present in the home, such as mice, can function as hosts.

Encountering Bed Bugs

Program staff wear the appropriate PPE (Personal Protective Equipment) including gloves, safety glasses, and/or shoe coverings during all site visits. If bed bugs are identified during site visits, Program staff should discretely gather their work tools and products and take them out of the house or apartment. Once removed from the area the items should be visually inspected for bed bugs. Items that contain bed bugs should be sealed in a garbage bag and removed for further inspection with his/her supervisor. The Installation Contractor contacts the Company Program Manager to notify staff of the situation.

Customer Notification:

If bed bugs are identified, the lead staff at the site notifies the homeowner or property manager, letting the customer know what was found, where the discovery was made, that belongings were removed and Program staff will not be returning to the home or unit.

Program Staff Communication:

The lead staff on site contacts the Company Program Manager after the customer is notified of the situation and next steps are identified. In the event a site visit (assessment, installation or inspection) is discontinued, the Installation Contractor will discuss possible next steps with the homeowner or property manager, including rescheduling a visit after steps are taken to remedy the situation. Prior to rescheduling, the homeowner or

property manager will need to provide proof that a professional pest control service was used to properly treat the area.

The Company Program Manager will notify the utilities of the situation and next steps, including rescheduling the visit should the situation be remedied.

6.5 Circuit Breaker Lockout Procedure

Proper circuit breaker lockout procedure is to be followed during installation of a measure that includes exposure to the electrical interface (110VAC or higher) of HVAC equipment, when the power switch to the device has been moved to the "Off" position and power to that device is still present after being tested with a current tester. This may occur during installation of a programmable thermostat, furnace or other electrically powered device.

The following steps are taken at the breaker box or service panel:

- Identify the proper breaker that is to be powered down. If the proper breaker cannot be identified, the work may not continue.
- Move the breaker switch to the “off” position.
- Place the lockout device over the desired breaker switch. If the lockout device is not compatible with the panel components and cannot be safely secured, the installation and all work related to the measure installation may not continue. The property may provide a lockout device that is compatible, but it must be carefully inspected and tested as functional.
- Tighten the set screw securely with a screwdriver.
- Attach the padlock.
- Verify that the lockout device secures the breaker switch in the “Off” position.

Prior to commencing work, the power to the device (e.g., thermostat, furnace, etc.) must be tested with a current tester and it must be confirmed that power has been successfully shut off.

Once the measure has been fully installed, and all safety protocols followed to restore any electrical panels to their original condition, the lockout/tag out device may be removed:

- Remove the padlock from the breaker switch disconnect
- Loosen the set screw with a screwdriver
- Remove the lockout device from the breaker switch
- Move the breaker switch to the "On" position

6.6 Whole House Fan Policy

When performing work in the attic of a home that contains a functioning whole house fan, ensure the fan is not operable while performing work in the home. Once work is complete in the home, the Contractor will enable the use of the fan.

- Educate the customer not to use the fan while work is being performed

If the customer wishes to have the whole house fan removed, the Contractor will remove the fan and permanently seal the area.

6.7 Fire Prevention and Mitigation

To ensure safety, we have established comprehensive Fire Prevention and Mitigation protocols for jobs including drilling and fire extinguisher accessibility:

Visual Inspection:

1. Before beginning any drilling or construction work perform a self-check to ensure that you are alert and wearing all necessary personal protective equipment. It is imperative to conduct a visual inspection to identify potential fire hazards. Look beyond the surface; assess what lies behind walls, ceilings, and floors. Check for nearby outlets that could pose wiring/electrical hazards and determine if there is masonry behind drywall plumbing (gas or water), among other considerations. Tools that can aid giving you this visibility are bore scope cameras and stud scanners.
2. When drilling a hole, check the location with a stud finder to avoid drilling through studs and beams where possible. If drilling through wood, take breaks to allow drill bits to cool before continuing.
3. Stop drilling after each layer and re-inspect thoroughly. Utilize all your senses – observe, smell, and feel for any abnormal heat or warmth. Do not proceed until you are certain there is no fire risk. This step ensures that any potential hazards are identified and addressed promptly.
4. Upon completing drilling, conduct a full inspection of the hole to ensure there are no signs of smoke, sparks, or smoldering. Utilize all your senses – observe, smell, and feel for any abnormal heat or warmth. Do not proceed until you are certain there is no fire risk.

Fire Extinguisher Accessibility:

1. It is imperative to have a plan in the event of a fire. Ensure fire extinguisher and fire alarm locations on the floor where work is being conducted are known. A fire extinguisher should be accessible in ~~If you carry a fire extinguisher, take it with you to~~ the immediate area in which you will be performing work. In the event of a fire, immediate access to extinguishing equipment can make a crucial difference in preventing the escalation of fire hazards.
    a. Recognize that during emergencies, time is of the essence. In the event of a fire outbreak on a higher floor, returning to a vehicle to retrieve a fire extinguisher may not be feasible. Having extinguishers in the immediate area in which you will be performing work ensures prompt response to fire emergencies.

Equipment Usage:

1. Always utilize the correct equipment for the task at hand. Using inappropriate or faulty equipment can increase the risk of fire hazards.
2. Regularly inspect and maintain all equipment to ensure it is in proper working condition. Faulty equipment not only compromises efficiency but also poses significant safety risks.

7. Deferral or Walk Away Policy

Should Installation Contractor encounter any of the following conditions they must be resolved by the Program or the homeowner for Installation Contractor to complete assessments, measure installations or inspections. The measures (both energy efficiency and health and safety) identified through the assessment should still be entered into the DCT should the project be able to proceed in the future. Resolvable issues prohibiting energy

efficiency work (i.e., health and safety issues exceeding $1500) should be reported to the customer immediately using the **Hazardous Condition Form (Appendix I)** where the issues to be resolved are outlined and the customer should be encouraged to reach out for energy efficiency services once those issues are resolved. If the home is located in Cook County and has health and safety issues exceeding the $1,500 threshold, but less than $14,000, they may qualify to use grant funds from the Federal Home Loan Bank (FHLB) Helping Homes program. Please share project details of these homes with the Retrofits PM to see if they qualify and, if so, work through the process to use grant funds.

- The customer has known health conditions that prohibits the installation of insulation and/or other weatherization materials.
- The building structure or its mechanical systems, including electrical and plumbing, are in such a state of disrepair that conditions prohibit weatherization.
- The house has sewage or other sanitary problems that would further endanger the client and weatherization installers if weatherization measures were installed.
- The house has been condemned or electrical, heating, plumbing, or other equipment has been "red tagged" by local or state building officials or utilities.
- Moisture or site drainage problems are so severe they cannot be resolved under existing allowable health and safety measures or with minor repairs.
- The client is uncooperative, abusive, or threatening to the crew, Installation Contractors, auditors, inspectors, or others who must work on or visit the house.
- The extent and condition of lead-based paint in the house would potentially create further health and safety hazards for the client or workers and could not be corrected with weatherization.
- Illegal activities are being conducted in the dwelling.
- Mold and Moisture problems are so severe they cannot be resolved or repaired within the health and safety limit.
- Areas cluttered or obstructed to such an extent that workers do not have access to home areas where assessment, weatherization work, or final inspection need to be performed.
- Pest Infestation, if the pests cannot be removed and pose a health and safety risk to the workers.
- Homes that contain products containing air pollutants, flammable liquids, and other volatile organic compounds that may pose a threat to workers may be deferred. Clients should be educated about the dangers of these products and informed that the products must be removed before work can commence.
- The presence of a pet that prohibits access or disturbing the site visit.
- Property is for sale, or in the process of being sold or property is in the process of legal foreclosure. Should Contractor become aware of either of these they should refer customer to Resource Innovations Program Manager for explanation work not proceeding. All work would stop immediately, and project would be closed out.
- Any other health or safety condition present or created by, exacerbated by, and cannot be corrected by the delivery of services, or where the Program representative feels personally at risk or uncomfortable.

9. Intellectual Property

Please see the following paragraph about intellectual property and how it relates to this program.

Contractor (and its subcontractors, consultants, or affiliates, as applicable) shall retain all right, title and interest in and to any and all works, submissions, designs, applications, know-how, expressions, inventions, ideas, discoveries, software, technology, information, materials, improvements, or developments as well as all copyrights, patents or trademarks thereof that are (i) developed or acquired by Contractor independently of

work under the Contract or (ii) in existence as of the commencement of the Services, including modifications and improvements thereto ("Contractor IP") that are incorporated or embedded in Deliverables, and including all third party works and products which Contractor is free to license. If Contractor IP is utilized to prepare the Deliverables, Contractor hereby grants Company and Client a royalty-free, irrevocable, worldwide, nonexclusive, perpetual license, to use, disclose, perform and display Deliverables internally. Notwithstanding any other provision of the Contract, Contractor retains ownership of all such Contractor IP, and subject to any applicable nondisclosure obligation, retains the right to use, disclose, reproduce, sublicense, modify, prepare derivative works from, perform and display its knowledge, experience and know-how, including processes, ideas and techniques developed or improved by Contractor in the course of performing the Contract. For the avoidance of doubt, Company IP includes Company's workflow management system, POWERPATH, which it licenses from and is owned by AIQUEOUS.

Appendix A: Safety Plan

Appendix B: HES Steps of Service Field Manual

Appendix C: Data Collection Tool Manual and Webinar Presentation

Appendix D: Audit Data Collection Form

This form was created to ensure all required systems information and project data is collected at the time of the energy audit appointment. Energy audit forms created and used by the Installation Contractor are acceptable as long as they guide the field technician to collect the required data.

Appendix E: Customer Authorization Form

Appendix F: Final Inspection Form

Appendix G: QAQC Observation Form

Appendix H: Mechanical Replacement Decision Trees

Appendix I: Hazardous Conditions Form

Appendix J: Documentation and Photo Checklist

Appendix K: Customer Satisfaction Survey